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MILITARY SPECIFICATION
CARTRIDGE, DELAY: M87, ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay: M87, Assembly for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-rings
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS
Munitions Command

D10520088	- Cartridge, Delay: M87, Assembly
C8864168	- Carton, Packing, Ammunition for Cartridge, Delay: M87
C9214156	- Box, Packing, Ammunition for Cartridge, Delay: M87
IEL-10520088	- Inspection Equipment List for Cartridge, Delay: M87, Assembly

FSC 1377

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(Copies of specifications, standards, drawings, and publications, required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Assoc. 10 E. 40th Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies, shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.- The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter, or propellant, or derangement of components.

3.4 Primer sensitivity.- The primer shall fire when drop tested at an energy level of 26 inch-ounces in a sectioned cartridge, void of delay charge, igniter, or propellant charge.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.- The cartridge when tested in an M20 or M21 Initiator shall comply with the requirements specified in Table I.

Table I

Temperature	Pressure (psi) 1/ Min	Max	Delay Time, Seconds (see 6.2)
-65° to -70°F	600	1400	1.30 max
65° to 75°F	700	2200	1.0 ± 0.1
200° to 205°F	800	2200	0.80 min

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1/ Pressure values shall be producted in a 1 cubic-inch pressure station at the end of 15 feet of hose (size No. 4), in accordance with MS-28741.

3.6 Workmanship.- Workmanship shall be in accordnace with the requirements of the applicable drawings, referenced specifications, and the following:

3.6.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity, or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At he beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 cartridge cases and 23 retaining rings for protective finish tests, 50 cartridge cases for hardness tests, 50 cartridges for primer sensitivity tests, 5 cartridges for sealant tests, and 30 cartridges for ballistic tests, and tested in accordance with Table II, and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

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Table II

Paragraph Requirement	Paragraph Test	Test	Sample Size	Type of Test: Destructive	Type of Test: Nondestructive
3.2	4.4.1	Protective finish	23 Cartridge cases	X	-
			23 Retaining Rings	X	-
Dwg	4.4.2	Hardness	50 Cartridge Cases	X	-
Dwg	4.4.3	Airtightness	Each	-	X
3.3	4.4.4	Vibration	12 (from Ballistic Samples)	-	X
3.4	4.4.5	Primer Sensivity	50	X	-
Dwg	4.4.6	Sealant application	5 cartridges	X	-
3.5	4.4.7.1	Ballistics	30	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for testing shall consist of the following:

Table III

Temperature	Vibrated	Not Vibrated
-65° to -70° F	4	6
65° to 75° F	4	6
200° to 205° F	4	6

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- An ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process.

b. Propellant from not more than 1 lot.

c. Primers from not more than 1 lot.

d. Igniters from not more than 1 lot.

e. Delay elements from not more than 1 lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic), the criteria in Table IV shall be used. For total defects within a major or minot class, the criteria in Table V shall be used. All nonconforming material shall be rejected. The roughness comparison specimen prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber o-rings shall be in accordance with MIL-STD-413. Four power magnification (4X) may be used in performing visual examination for such defects as burr and foreign matter.

Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

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Table V

Total Allowable Defects per Sample Size

	Major		Minor	
	Accept	Reject	Accept	Reject
Sample Size for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Size for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Delay Element, M23, Drawing D10520090.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Mixed ammunition	Visual
3. Flash hole missing or obstructed	Visual
4. Marking missing, incorrect of illegible	Visual
5. Delay charge not visible through hole	Visual
MAJOR	
101. Fpreign matter	Visual
102. Protective finish inadequate or defective	Visual
MINOR	
None Defined	

4.3.2.2 Ring, Retaining, Drawing B8597235, covering a detail of Dwg D10520088.

CRITICAL	
1. Metal defective	Visual

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4.3.2.2 (cont)

Defect	Method of Inspection
MAJOR	
101. Concentricity of inside diameter with outside diameter	Gage
102. Diameter inside, max.	Gage
103. Diameter outside, min	Gage
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
107. Thickness	Gage
MINOR	
201. Diameter inside, min	Gage
202. Diameter outside, max	Gage
203. Surface finish improper	Visual
204. Total length	Gage
205. Radius missing or incorrect	Visual

4.3.2.3 Case, Cartridge, Drawing C10519996, covering a detail of Dwg D10520088.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of cartridge head with body	Gage
102. Straightness of base	Gage
103. Thickness of wall, min	Gage
104. Thickness of base, min	Gage
105. Protective finish inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual
108. Diameter of Lead, min	Gage
MINOR	
201. Diameter of head, max	Gage
202. Diameter of body	Gage
203. Total length	Gage
204. Location of datum diameter	Gage
205. Radius missing or incorrect	Visual
206. Surface finish improper	Visual
207. Thichness of wall, max	Gage
208. Thickness of base, max	Gage

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4.3.2.4 Assembly, Cartridge, Delay, M87, Drawing D10520088.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray 1/ and Visual
2. Improper assembly	X-ray 1/ and Visual
3. Metal defective	Visual
4. Marking missing, incorrect, or illegible	Visual
5. Mixed ammunition	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of cartridge head recess, min.	Gage
103. Diameter of body, max	Gage
104. Diameter of cartridge head, max.	Gage
105. Depth of cartridge head recess	Gage
106. Location of datum diameter	Gage
107. Cartridge profile failure	Gage
108. Firing pin or retaining cup above flush with face of head	Visual
109. Foreign matter	Visual
MINOR	
201. Total length, max	Gage
202. Fold, wrinkle, dent, buckle, bulge or scratch	Visual
203. Varnish missing or incorrect	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.5 Carton, Inner Packing for Cartridge, Delay, M87, Drawing C8864168.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual

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4.3.2.5 (Cont) MINOR

Defect	Method of Inspection
204. Heat sealer incomplete, loose, or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect of illegible	Visual

4.3.2.6 Box, Packing for Cartridge, Delay, M87, Drawing C9214156.

CRITICAL
None defined

MAJOR
None defined

MINOR

201. Contents loose	Manual and Visual
202. Hardware missing, loose, broken, or improperly engaged	Visual
203. Strapping missing, loose, broken, or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect, or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed, or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Protective finish.- Twenty-three cartridge cases, and 23 retaining rings, shall be selected for this test. Failure of any component to comply with the requirements of 3.2, shall be cause for rejection of the lot.

4.3.3.2 Hardness.- Fifty units from each heat-treated lot shall be selected for this test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.3 Airtightness.- Each cartridge shall be subjected to the airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

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4.3.3.4 Vibration.- Twleve of the 21 assembled x-rayed cartridges of the initial production sample shall be subjected to vibration as in 3.3. After vibration test, the 12 vibrated cartridges shall be x-rayed, as directed in Note 1/, 4.3.2.4. Evidence of any disintegration of the primer, propellant, igniter, delay charge, or derangement of any component, shall be cause for rejection of the lot, as indicated in 3.3. When a change occurs in the lot of primers, propellant, igniters, or delay charge, after the initiation of production, 12 assembled cartridges from the first production lot shall be tested, in accordance with this paragraph.

4.3.3.5 Sealant application.- Five cartridges from each production lot shall be utilized for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.6 Primer sensitivity.- A sample of 50 assembled cartridges from the initial production sample, and 25 assembled cartridges from each subsequent production lot, shall be tested for sensitivity. Failure of any cartridge to comply with the requirements of 3.4, shall be casue for rejection of the lot.

4.3.3.7 Functional.- The following tests shall be performed by the Government:

4.3.3.7.1 Ballistics.- The 30 assembled cartridges from each production lot shall be grouped and temperature conditioned, in accordance with 4.4.7.1.1 and Table VI. Failure of any cartridge to comply with the requirements of 3.5, shall be cause for rejection of the lot.

Table VI

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
200° to 205°F	10

4.3.4 Inspection equipment.- The examination and tests shall be performed using equipment listed on IEL-10520088.

4.4 Test methods and procedures.

4.4.1 Protective finish.- Test proceudres for protective finish shall be in accordance with MIL-C-60539. All anodized components shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Hardness.- A hardness testing machine shall be used to determine compliance with drawing requirements.

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4.4.3 Airtightness.- The method for determining airtightness shall be by use of the equipment listed on IEL-10520088.

4.4.4 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.5 Primer sensitivity.- Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge around its periphery 1 inch, maximum, above the base. Drop test 50 sectioned cartridges from the production sample, and 25 sectioned cartridges from each production lot (all void of charged delay body and powder charges) at 13 inches, using a steel ball 15/16 inch in diameter, weighing 1.94 ounces, plus or minus 0.02 ounce, in a primer sensitivity machine with a cartridge holder approved by the procuring activity.

4.4.6 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length (360°) of the joining surfaces at the junction of the o-ring and case sidewall and that the propellant cavity is free of sealant.

4.4.7 Functional.

4.4.7.1 Ballistics.- Cartridges of the sample shall be grouped and temperature conditioned in accordance with 4.4.7.1.1, and rigidly supported in the test fixture. The sample shall be tested within five minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.7.1.1.

4.4.7.1.1 Temperature conditioning.- A standard inert cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8864168 and C9214156.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

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6.2 Delay time is defined as the time lapse between the firnig of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Supercession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2443, dated 7 March 1961.

Custodian:

Army -MU

Preparing Activity;

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